

# DP Computer Science: A 1.1.9 Cloud Computing

## 1. Core Concepts & Learning Objectives

### Learning Intentions

By the end of this lesson, students will be able to:

1. Describe what **cloud computing** is: renting computing resources over the internet and paying for what you use
2. Describe what each of the three service models (**SaaS**, **PaaS**, **IaaS**) provides, and **who manages what** in each
3. Describe how the degree of **control and flexibility** affects resource management and resource availability across the models
4. Given a real scenario, describe **which model fits and why**

**Command Term Note:** The lesson is held at **Describe**—a detailed account. Cost-benefit and pros-and-cons work has been cut because they belong to an Evaluate command term this statement does not carry.

### Key Vocabulary (Tier 3)

| Term                               | Definition                                                                                          |
|------------------------------------|-----------------------------------------------------------------------------------------------------|
| Cloud computing                    | Renting computing resources over the internet and paying for what you use                           |
| SaaS (Software as a Service)       | Ready-to-use applications delivered over the internet                                               |
| PaaS (Platform as a Service)       | A platform for developing and deploying applications, with the provider managing the infrastructure |
| IaaS (Infrastructure as a Service) | Rented fundamental resources (virtual servers, storage, networking) that the customer builds on     |
| Scalability                        | The ability to increase or decrease resources to match demand                                       |
| Resource pooling                   | One provider serving many customers from shared physical resources                                  |

### Pedagogical Focus

- **Research Skills (ATL):** Students consciously practise research skills. Each student or pair takes one real service or organisation, locates a credible source, and records who manages which layer of the stack, noting where the information came from and whether the source can be trusted.
- **Global Contexts (ATT):** The lesson is driven by real organisations rather than by an abstract feature list. Open with three that make different choices—a school on SaaS, a small team on PaaS, a streaming firm on IaaS—and let the differences in who manages what surface from what each organisation is trying to do. The return brings in the global dimension: the data centres behind the service and their environmental cost, described rather than debated.

## 2. Lesson Sequence (One-Hour Block)

### Part 1: Do Now (Retrieval) — ~10%

**Activity:** Three short recall items from earlier taught content:

| Question                                                                         | Source   |
|----------------------------------------------------------------------------------|----------|
| 1. What is secondary storage, and why does a computer need it?                   | [A1.1.7] |
| 2. What does a server do, and what part does the CPU play in it?                 | [A1.1.1] |
| 3. Where are GPUs used at scale, for example in AI training inside data centres? | [A1.1.2] |

Part 2: Main Sequence — ~60%

Phase A: Anchor — Three Organisations (~8 mins)

Present three organisations with different needs:

| Organisation                                                                                           | Need                                                     |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|  School                | Wants email and shared documents for staff and students  |
|  Two-Person Team       | Launching a web app; needs to develop and deploy quickly |
|  Streaming/Retail Firm | Running a custom system at scale                         |

Task: For each organisation, ask:

- "What does the organisation want to do?"
- "How much of the system would it want to run itself?"

Students record a first guess before any theory is given.

Phase B: Describe Through the Anchors (Worksheet-Led) (~18 mins)

Build the Three Service Models:

text

SaaS

Ready-to-use applications

User manages NOTHING – just uses the app

Examples: Gmail, Google Docs, Dropbox

PaaS

Platform for developing and deploying

User manages application code and configuration

Provider manages infrastructure (servers, OS)

Examples: Google App Engine, Heroku

IaaS

Fundamental resources (servers, storage, net)

User manages everything from OS upward

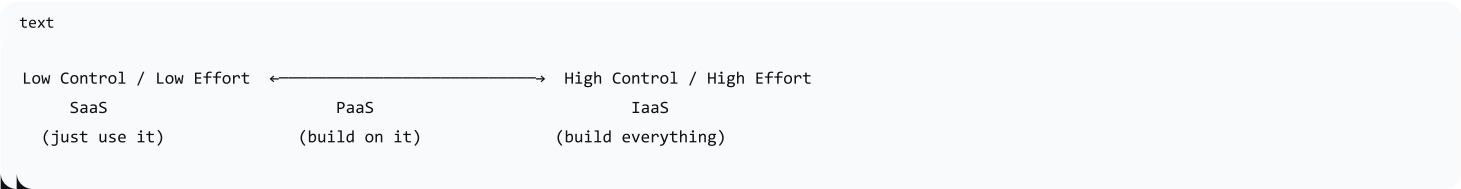
Provider manages physical hardware and virtualisation

Examples: AWS EC2, Google Compute Engine

The "Who Manages What?" Stack:

| Layer            | SaaS     | PaaS     | IaaS     |
|------------------|----------|----------|----------|
| Application      | Provider | User     | User     |
| Data             | Provider | User     | User     |
| Runtime          | Provider | Provider | User     |
| Operating System | Provider | Provider | User     |
| Virtualisation   | Provider | Provider | Provider |
| Servers/Storage  | Provider | Provider | Provider |
| Networking       | Provider | Provider | Provider |

Control and Flexibility Scale:



**Key Insight:** As control increases, management responsibility and technical expertise requirements also increase.

Where the Common IaaS Misconception is Corrected:

*"IaaS is **not** just a server in the cloud. The customer must manage the OS, security patches, and application stack. The provider only manages the physical hardware."*

Phase C: Return with the Global Dimension (~4 mins)

Come back to the anchors—what sits behind the service:

| Aspect             | Description                                                            |
|--------------------|------------------------------------------------------------------------|
| Data Centres       | Massive facilities housing thousands of servers                        |
| Data Location      | Data may be stored in specific countries with different jurisdictions  |
| Environmental Cost | Data centres consume enormous amounts of electricity (cooling + power) |

**Keep this descriptive; it feeds the research strand rather than a debate.**

Part 3: Deliberate Practice — ~20%

Activity A — Categorise Real Services (Pairs):

| Service           | Categorise<br>(SaaS/PaaS/IaaS) | Who Manages What?                                      |
|-------------------|--------------------------------|--------------------------------------------------------|
| Gmail             | SaaS                           | Provider manages everything; user just uses email      |
| Google App Engine | PaaS                           | User manages app code; provider manages infrastructure |

| Service | Categorise<br>(SaaS/PaaS/IaaS) | Who Manages What?                                                  |
|---------|--------------------------------|--------------------------------------------------------------------|
| AWS EC2 | IaaS                           | User manages OS and everything above;<br>provider manages hardware |
| Dropbox | SaaS                           | Provider manages storage service; user just<br>syncs files         |
| Heroku  | PaaS                           | User deploys code; provider manages<br>runtime and servers         |

#### Activity B — Scenario Fit (Worksheet Parts 5 and 6):

For each short scenario, describe which model fits and why:

| Scenario                                                                                | Best Fit | Why?                                                                 |
|-----------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------|
| A school needs staff email                                                              | SaaS     | No technical expertise; ready-to-use;<br>provider manages everything |
| A startup building a new web app<br>wants to deploy quickly without<br>managing servers | PaaS     | Provider manages infrastructure; team<br>focuses on code             |
| A large retail firm needs custom<br>virtual servers with full control over<br>the OS    | IaaS     | Maximum control and flexibility; firm<br>has IT team to manage       |
| A small business wants to store and<br>share files                                      | SaaS     | Ready-to-use storage; no<br>infrastructure to manage                 |

## Part 4: Exit Check — ~10%

Two-Line "Who Manages What" for One Named Service:

| "For [*named service*], the provider manages \_\_\_\_, while the customer manages \_\_\_\_\_. This makes it a \_\_\_\_ model."

Example:

| "For Google App Engine, the provider manages the servers and runtime environment, while the customer manages the application code and configuration. This makes it a PaaS model."

| Readable in minutes and tied directly to the statement.

## Homework (Consolidation)

| Task     | Description                    |
|----------|--------------------------------|
| Read     | A1.1.9 eBook page              |
| Complete | Ten-question quiz on the eBook |

| The quiz doubles as retrieval evidence to open the next lesson.

## 3. Summary: The Three Service Models

## SaaS — Software as a Service

| Feature          | Description                                                   |
|------------------|---------------------------------------------------------------|
| What it provides | Ready-to-use applications over the internet                   |
| User manages     | Nothing — just uses the application                           |
| Provider manages | Everything (application, data, runtime, OS, servers)          |
| Control          | Minimal                                                       |
| Examples         | Gmail, Google Docs, Dropbox, Microsoft 365                    |
| Best for         | Users who want applications without management responsibility |

## PaaS — Platform as a Service

| Feature          | Description                                              |
|------------------|----------------------------------------------------------|
| What it provides | Platform for developing and deploying applications       |
| User manages     | Application code and configuration                       |
| Provider manages | Infrastructure (servers, runtime, OS, networking)        |
| Control          | Medium                                                   |
| Examples         | Google App Engine, Heroku, AWS Elastic Beanstalk         |
| Best for         | Developers who want to focus on code, not infrastructure |

## IaaS — Infrastructure as a Service

| Feature          | Description                                                         |
|------------------|---------------------------------------------------------------------|
| What it provides | Fundamental resources (virtual servers, storage, networking)        |
| User manages     | Everything from OS upward (OS, security patches, application stack) |
| Provider manages | Physical hardware and virtualisation                                |
| Control          | Maximum                                                             |
| Examples         | AWS EC2, Google Compute Engine, Microsoft Azure VMs                 |
| Best for         | Organisations with IT teams needing full control                    |

## Comparison Table

|                  | SaaS             | PaaS                 | IaaS                      |
|------------------|------------------|----------------------|---------------------------|
| What you get     | Ready-to-use app | Development platform | Virtual infrastructure    |
| Your control     | None             | App code + config    | OS and everything above   |
| Provider manages | Everything       | Infrastructure       | Hardware + virtualisation |

|                 | SaaS  | PaaS              | IaaS    |
|-----------------|-------|-------------------|---------|
| Effort required | Low   | Medium            | High    |
| Example         | Gmail | Google App Engine | AWS EC2 |

## 4. Differentiation & Extension

### Support (Scaffolds That Can Be Removed)

| Support Strategy                  | Description                                                                |
|-----------------------------------|----------------------------------------------------------------------------|
| Sentence Frame                    | For the written categorisation and scenario descriptions                   |
| Partially Completed Stack Diagram | Layer names filled in; "manages what" column blank                         |
| Curated Service List              | For the research task; withdraw once students can find and judge their own |

### Challenge (Deeper or Wider, Held at Describe)

| Challenge Strategy             | Description                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One Organisation, Three Models | Describe how a single organisation can use all three models at once (e.g., a retailer using SaaS for email, PaaS for web app, IaaS for big data processing) and say who manages what in each |
| Specific Provider Research     | Research a specific provider's named service and describe exactly where the management boundary falls                                                                                        |

Keep extension at description, not evaluation.

## 5. Engagement Activities (Supplementary)

*If time allows, these can replace or supplement the main sequence:*

| Activity                       | Description                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| "Cloud Brainstorming"          | Students brainstorm everything they associate with the word "cloud"                                                      |
| "Everyday Cloud" Discussion    | Discuss how students already use cloud services (social media, online storage, streaming)                                |
| "Cloud Service Scavenger Hunt" | Groups categorise popular cloud services as IaaS, PaaS, or SaaS                                                          |
| "Cloud Service Analogy"        | Groups present differences using analogies (IaaS = renting land, PaaS = renting an apartment, SaaS = staying in a hotel) |
| "Cloud Consulting" Role-Play   | Groups role-play a consultation to determine the most suitable cloud solution for a business scenario                    |

| Activity              | Description                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| "Cloud Ethics Debate" | Discuss ethical implications: data privacy, security concerns, environmental impact of data centres |

## 6. Learning Objectives (Summary)

### Knowledge and Understanding

- Define cloud computing and its key characteristics
- Identify and differentiate between SaaS, PaaS, and IaaS
- Explain each service model's different levels of control, flexibility, and responsibility
- Describe real-world scenarios and applications of each cloud service model

### Skills

- Analyse different computing needs and recommend appropriate cloud service models
- Evaluate trade-offs between different cloud service models regarding cost, control, flexibility, and security
- Research and compare different cloud providers and their offerings

### ATL Skills

| Skill                | Description                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------|
| Communication        | Effectively communicate understanding through discussions and written explanations             |
| Collaboration        | Work collaboratively to analyse scenarios, research cloud services, and design cloud solutions |
| Critical Thinking    | Analyse advantages/disadvantages; evaluate suitability for various applications                |
| Information Literacy | Use online resources to deepen understanding of cloud technologies and trends                  |
| Reflection           | Consider impact of cloud computing on businesses, society, and personal use                    |
| Transfer             | Apply understanding to new and emerging technologies and services                              |

## 7. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A 1.1.9 with quiz)
- **Worksheet:** Stack diagram + categorisation + scenarios
- **Worksheet Answers**

### Author's Reflection

*"I have anchored this in organisations rather than in the abstract SaaS, PaaS, and IaaS stack, because the models only make sense once you can see who is left holding which part of the system: a school that patches nothing against a streaming firm whose engineers patch everything. The command term is*

*Describe, so I have stopped at that level.*

*Students say what each model provides and who manages what, and I have kept out the cost-benefit and pros-and-cons work the old lesson drifted into, which belongs to an Evaluate command term this statement does not carry.*

*The trade-off I am conscious of is that cloud almost begs to be evaluated, so the discipline here is to let the scenarios do the describing and stop short of a verdict. The eBook and its quiz do the consolidation at home, which keeps class time on the contexts and the research."*